Research Statement

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I study *higher category theory* and in particular higher topos theory. My work has been focused on developing a theory of *elementary higher toposes*, which at the same time generalizes elementary toposes and higher toposes. Intuitively, it should be thought of as a higher category where we can use the tools we have developed to study spaces or, in other words, where we can use the tools and results of algebraic topology. It is my long term goal to use the theory I have developed to gain a better understanding of the higher category of spaces.

0. From Classical to Derived: Grothendieck Toposes and Higher Toposes

Topos theory was developed for classical categories by Grothendieck and Bourbaki in order to define sheaves on a category and tackle problems in algebraic geometry [GJ72]. Around the same time, Lawvere axiomatized the category of sets, characterizing categories whose objects behave like sets [La65]. After joining forces with Tierney they generalized Grothendieck toposes to so called *elementary toposes*, that can study sets from a categorical perspective [Ti73]. Using the theory of elementary toposes Lawvere and Tierney classified set theories. As an application they showed the independence of the continuum hypothesis from ZFC [Ti72].

Over time algebraic geometers realized they needed methods to preserve higher level information. This development gave rise to *derived algebraic geometry*, which uses simplicial commutative rings [Lu12]. The study of derived algebraic geometry meant that Grothendieck toposes were not sufficient, which required the introduction of *higher categories*. Higher categories generalize classical categories and are used to study a collection of objects that has higher (or derived) data [Be10]. They first arose in algebraic topology where we study topological spaces with homotopical data. Thus topos theory had to be redeveloped for higher categories in order to be applied to derived algebraic geometry, which was done by Lurie. He developed the necessary higher categorical tools and defined *higher topos theory* [Lu09A].

My research focuses on generalizing elementary toposes to a higher categorical analogue, *elementary higher toposes*. My work sets up the necessary machinery to develop and study such a theory. I plan to use this theory in other areas such as algebraic topology and homotopy type theory.

1. Main Results

Here is my definition and my main results about elementary higher toposes.

Definition 1.1 ([RaD])**.** An elementary higher topos (EHT) $\mathcal{E}$ is a higher category that satisfies the following conditions

1. $\mathcal{E}$ has finite limits and colimits.
2. $\mathcal{E}$ has a subobject classifier (Definition 2.1).
3. For each morphism f there exists an object classifier $\mathcal{U}_\bullet$ such that f is in the image of $\mathcal{Y}_{\mathcal{U}_\bullet}$ (Definition 4.4).

Theorem 1.2 ([RaD])**.** *If $\mathcal{E}$ is an EHT and c is any object, then $\mathcal{E}_{/c}$ is also an EHT.*

Remark 1.3. In the classical setting this result is known as the *fundamental theorem of topos theory* ([Fr72]) as it indicates the persistence of topos theoretic phenomena.

Theorem 1.4 ([RaD])**.** *Every elementary higher topos is locally Cartesian closed.*

Remark 1.5. This is a surprisingly strong result as any other definition of an elementary topos, classical or higher, assumes some part of the conditions of a Cartesian closed category. However, the closure condition follows from the strengthened notion of an object classifier introduced here (Definition 4.4).

Proposition 1.6 ([RaD])**.** *Every elementary higher topos satisfies descent.*

Remark 1.7. Descent is the categorification of the sheaf condition and thus plays a central role in higher topos theory ([Re05, Section 6.5], [Lu09A, Section 6.1.3]) to the extent that we can even take it as part of the definition of a higher topos. In the context of an EHT we can deduce it by working with a stronger notion of an object classifier.

Corollary 1.8. *A presentable higher category is a higher topos if and only if it is an elementary higher topos.*

Remark 1.9. This implies that every higher topos is also an elementary higher topos. So, in particular the category of spaces is an elementary higher topos.

2. Subobject Classifiers vs. Object Classifiers

Let us start by reviewing the definition of an elementary topos.

Definition 2.1. An elementary topos is a locally Cartesian closed category $\mathcal{C}$ with subobject classifier Ω. A subobject classifier is any object that represents the functor $Sub(-)$, the functor that takes each object to the set of isomorphism classes of subobjects (monomorphisms over the given object).

The reaon we only classify monomorphisms is that more general classes of maps can have *non-trivial automorphisms*, which cannot be preserved by hom sets. In the realm of higher category theory, however, such restrictions don't exist, as objects have *mapping spaces* rather than mapping sets and we can classify automorphisms using homotopies of that space. Concretely we have following definition.

Definition 2.2. Let $\mathcal{C}$ be a higher category with finite limits. There is a functor

$$\mathcal{C}_{/-} : \mathcal{C}^{op} \to \mathcal{C}\mathrm{at}_\infty$$

that takes each point c to the over category $\mathcal{C}_{/c}$ (the higher category of all objects over c).
This functor can be post-composed to give a functor

$$(\mathcal{C}_{/-})^{core} : \mathcal{C}^{op} \to Spaces$$

which takes each point c to the maximal higher subgroupoid (called the *core*) of $\mathcal{C}_{/c}$.

Definition 2.3. An object $\mathcal{U}$ is an *object classifier* if there is an embedding from the representable functor

$$\mathcal{Y}_{\mathcal{U}} \hookrightarrow (\mathcal{C}_{/-})^{core}$$

such that the set of maps in the image are closed under finite limits and colimits.

Remark 2.4. The reason we consider an embedding rather than an actual equivalence is to avoid set theoretical issues.

This definition allows us to capture objects and their automorphisms which is a big step forward, however we did restrict to the core, which means we lost all of the category theory along the way. We would like to represent the functor $\mathcal{C}_{/-}$ rather than restricting to $(\mathcal{C}_{/-})^{core}$, however, it is impossible to do that with one object. We need to develop a theory of representability for functors that take value in higher categories. This leads us to study *complete Segal objects* and *representable Cartesian fibrations*.

3. Complete Segal Objects

Complete Segal Spaces (CSS) are a model for a higher category and have been developed by Charles Rezk [Re01]. One fascinating aspect about them is that the definition does not rely on any specific property of spaces, except for the fact that spaces have finite limits, allowing for a ready generalization.

Definition 3.1 ([RaC])**.** Let $\mathcal{C}$ be a higher category with finite limits. A *complete Segal object* is a simplicial object $X_\bullet : \Delta^{op} \to \mathcal{C}$ satisfying following conditions

1. *Segal condition*: For $n \geq 2$ the following map in an equivalence in $\mathcal{C}$.

$$X_n \to X_1 \underset{X_0}{\times} ... \underset{X_0}{\times} X_1$$

2. *Completeness condition*: The following map is an equivalence in $\mathcal{C}$.

$$X_0 \to X_3 \underset{(X_1 \times X_1)}{\times} (X_0 \times X_0)$$

A complete Segal object gives us a model for an internal *higher category object*. In particular, it comes with a notion of objects $x \in X$, morphisms $map_X(x, y)$, composition and equivalences. Thus it satisfies many things that we already know hold for higher categories. In particular, I proved that an important result from [Re01] also holds in this setting.

Theorem 3.2 ([RaC])**.** *Let $F : X_\bullet \to Y_\bullet$ be a map of complete Segal objects. Then F is a level-wise equivalence if and only if*

1. *Fully Faithful: For any two objects $x_1, x_2 \in X$ the induced map is an equivalence in $\mathcal{C}$.*

$$map_X(x_1, x_2) \to map_Y(Fx_1, Fx_2)$$

2. *Essentially Surjective: For any object $y \in Y$ there is an object $x \in X$ such that Fx is equivalent to y.*

One thing that we know about objects in a category is that we can build representable functors out of them. Having defined complete Segal objects, we would like to show that we can now build functors valued in higher categories out of them. However, this is quite tricky. In the the world of higher categories it is difficult to study functors, because of all the higher coherence conditions that arise when trying to analyze the functoriality condition. One way to tackle this problem is to model functors by certain fibrations and then study those fibrations instead.

4. Cartesian Fibrations

There are various fibrations that arise in the world of higher category theory. In particular, we study functors valued in the category of spaces through the theory of *right fibrations* ([Lu09A, Chapter 2]) and functors valued in higher categories through *Cartesian fibrations* ([Lu09A, Chapter 3]). In order to get a functioning theory of representability for functors valued in higher categories, I revisited some of the foundations on Cartesian fibrations in the context of bisimplicial spaces. The results can be distilled to the following theorem.

Theorem 4.1 ([RaA], [RaB])**.** *Let $\mathcal{C}$ be simplicial space. There is a unique simplicial left proper model structure on bisimplicial spaces over $\mathcal{C}$, $(ss\mathcal{S}_{/\mathcal{C}})^{Cart}$, called the Cartesian model structure, such that*

1. *The fibrant objects are the Cartesian fibrations over $\mathcal{C}$.*
2. *A map $\mathcal{D} \to \mathcal{E}$ over $\mathcal{C}$ is a weak equivalence if*

$$map_{ss\mathcal{S}_{/\mathcal{C}}}(\mathcal{E}, \mathcal{K}) \to map_{ss\mathcal{S}_{/\mathcal{C}}}(\mathcal{D}, \mathcal{K})$$

   is an equivalence for every Cartesian fibration $\mathcal{K} \to \mathcal{C}$.

3. *A map $\mathcal{D} \to \mathcal{E}$ over $\mathcal{C}$ is a Cartesian equivalence if and only if*

$$\mathcal{C}_{c/} \times_{\mathcal{C}} \mathcal{D} \to \mathcal{C}_{c/} \times_{\mathcal{C}} \mathcal{E}$$

   is a categorical equivalence for every c in $\mathcal{C}$. Here $\mathcal{C}_{c/}$ is a coCartesian fibrant replacement of c.

4. *If $\mathcal{C}$ is also a Segal space, then for any object $c \in \mathcal{C}$, the map*

$$* \longrightarrow \mathcal{C}_{/c}$$

   is a Cartesian equivalence over $\mathcal{C}$. Here $\mathcal{C}_{/c}$ is the Segal space of objects over c.

5. *A map $f : \mathcal{C} \to \mathcal{D}$ gives a Quillen adjunction*

$$(ss\mathcal{S}_{/\mathcal{C}})^{Cart} \underset{f^*}{\overset{f_!}{\rightleftarrows}} (ss\mathcal{S}_{/\mathcal{D}})^{Cart}$$

   which is a Quillen equivalence if f is a categorical equivalence.

6. *$ss\mathcal{S}_{/\mathcal{C}}$ is a localization of the complete Segal space (CSS) model structure on $s\mathcal{S}_{/\mathcal{C}}$.*
7. *Base change along Cartesian fibrations preserves CSS equivalences.*

Having developed a theory of Cartesian fibration, I could prove the following result on representable Cartesian fibrations.

Theorem 4.2 ([RaB])**.** *For every complete Segal object $X_\bullet$ in $\mathcal{C}$, there is a Cartesian fibration $\mathcal{Y}_{X_\bullet}$ that models the functor which at point c takes the value of the complete Segal space*

$$map_{/\mathcal{C}}(c, X_0) \Leftarrow map_{/\mathcal{C}}(c, X_1) \Lleftarrow map_{/\mathcal{C}}(c, X_2) \Llleftarrow \cdots$$

Moreover, we say a functor $F : \mathcal{C}^{op} \to \mathcal{C}at_\infty$ is representable if it is equivalent to $\mathcal{Y}_{X_\bullet}$ for some $X_\bullet$.

This construction behaves similar to the usual representable presheaves. In particular we have a Yoneda lemma

Theorem 4.3 ([RaB]). *For two complete Segal objects $X_\bullet$ and $Y_\bullet$, there is an equivalence*

$$map_{\mathcal{C}}(\mathcal{Y}_{X_\bullet}, \mathcal{Y}_{Y_\bullet}) \xrightarrow{\simeq} map(X_\bullet, Y_\bullet)$$

Having set up a theory of representable Cartesian fibrations, we can finally give the following definition.

Definition 4.4 ([RaD]). A complete Segal object $\mathcal{U}_\bullet$ is an *object classifier* if there is an embedding

$$\mathcal{Y}_{\mathcal{U}_\bullet} \hookrightarrow \mathcal{C}_{/-}$$

such that the set of maps in the image are closed under finite limits and colimits.

This summarizes all the tools we need in order to define elementary higher toposes and prove the results stated in Section 2.

5. Future Work

My research goals fall into three broad categories. First, I want to further develop the theory of elementary higher toposes. Second, I want apply this theory to other related fields of mathematics. Third, I want to expand the theoretical machinery I have developed for higher categories and various fibrations.

Gain a better understanding of EHT:

1. *Colimits:* In an elementary topos the existence of finite colimits follows from the other axioms. It is unclear whether the same is true in the higher categorical setting. There is an ongoing project with Jonas Frey we address that problem. Concretely, our current work indicates that following result holds true.

   Conjecture 5.1. *Every locally Cartesian closed higher category with subobject classifier has an initial object and finite coproducts.*

   A further goal is to investigate if we deduce the existence of pushouts from the other axioms of an elementary higher topos.

2. *Comparing Definitions:* Shulman has another definition of an elementary higher topos [Sh17].

   Definition 5.2. *(Shulman)* An elementary higher topos is a locally Cartesian closed higher category with finite limits and colimits, and a subobject classifier such that for every map f there is exists an object classifier $\mathcal{U}$ that classifies f.

   Theorem 1.4 shows that my definition is stronger. The opposite direction follows from the following conjecture.

   Conjecture 5.3. *Let $\mathcal{E}$ be a locally Cartesian closed category with object classifier $\mathcal{U}$. Then there exists an object classifier $\mathcal{V}_\bullet$ (Definition 4.4) such that $\mathcal{V}_0$ is equialent to $\mathcal{U}$.*

3. *Localizations:* Localizations of higher toposes are difficult to characterize given the amount of data that is involved. However, in elementary toposes the subobject classifier allows us to define Lawvere-Tierney topologies classify all localizations. My plan is to use the object classifier in elementary higher toposes to generalize those topologies to the higher categorical setting.

4. *Univalence in an EHT:* One problem in an EHT is managing object classifiers as the definition only assumes existence and does not provide any concrete tools to handle them. In particular, this makes it difficult to define and study functors between EHTs. In [GK17] Gepner and Kock develop a theory of univalence for locally Cartesian closed presentable quasicategories. Their work relies on the existence of object classifiers, so there is good indication that it will generalize to EHTs. The hope is that a theory of univalent families would allow us to make sense of bounded local classes in an EHT which in turn can help us study object classifiers.

5. *Algebraic Topology:* As indicated in the introduction, an EHT should be a higher category where we can use the tools of algebraic topology. The hope is that we can make this precise by showing that classical results about spaces hold in an EHT. My first step towards that is to prove following conjecture

Conjecture 5.4. *The Blakers-Massey theorem holds in every EHT.*

This result has already been proven in a higher topos ([Re05] Proposition 8.16). The goal is to show that the proof there generalizes to EHTs using the fact that both satisfy descent.

Applications to other areas of Mathematics:

1. *Categorical Definition of Spaces:* The main goal of all of my work has been to develop an axiomatic definition for the category of spaces. I now have a reasonable conjectural definition and want to use it to built nonstandard models for spaces. My long term goal is to use such models to shed new light on the homotopy theory of spaces.

2. *Models of Homotopy Type Theory:* Homotopy type theory is a foundational approach to mathematics in which homotopy theory is automatically built into the constructions [Vo10]. While we do have some models for it ([KL12]), we certainly still lack a complete understanding. There is some evidence that elementary higher toposes give us all the models for homotopy type theory and I hope to pursue this classification.

3. *Universal Higher Toposes:* One of the idealistic goals of topos theory is constructing a topos out of a category in a universal manner. This might not be possible for every category, but even specific cases can yield valuable results. One important example is the motivic category [Ro12]. While the stabilization behaves very well, the unstable category is often hard to manage partly because it does not satisfy descent.

   One of the benefits of my approach to EHTs is that it focuses all the attention on the object classifier. Therefore there is an actual chance we can study how far a category is from being a topos. The hope is that we can use EHTs to find an actual topos that is still reasonably close to the current motivic category.

Developing Higher Categorical Tools:

1. *Modeling functors valued in (∞, n)-categories using fibrations:* The theory of Cartesian fibrations is a crucial part of the theory of $(\infty, 1)$-categories and has enabled us to do various constructions, such as limits, adjunctions and monoidal structures. Thus I want to generalize it to the context of (∞, n)-categories. Concretely, I want to be able to define fibrations that model functors

$$\mathcal{C}^{op} \to \mathcal{C}\mathrm{at}_{(\infty,m)}$$

   from the (∞, n)-category $\mathcal{C}$ to the higher category of (∞, m)-categories. This has been done for the case of $n = 2$, $m = 1$ using a generalization of simplicial sets, *scaled simplicial sets* [Lu09B]. However, it is difficult to further generalize that construction as it relies on having a working model of (∞, n)-categories that generalizes quasicategories.

   My approach to Cartesian fibrations relies on complete Segal spaces, which have been generalized to models of (∞, n)-categories by Bergner and Rezk [BR13], [BR14]. My plan is to use their constructions to generalize my Cartesian fibrations to fibrations that model the above functors.

2. *Right Properness and Categorical Equivalences:* The CSS model structure is not right proper, thus base change along fibrations does not preserve equivalences. On the other hand we know that this property holds for Cartesian fibrations. This leads to the following natural question.

   Question 5.5. *Can we characterize the class of all maps that preserve equivalences in the CSS model structure?*

References

[Be10] J.E. Bergner, *A survey of (∞,1)-categories*, The IMA Volumes in Mathematics and its Applications. Springer, 2010. pp. 69-83

[BR13] J. Bergner, C. Rezk. *Comparison of models for (∞, n)-categories, I.*, Geometry & Topology 17.4 (2013): 2163-2202.

[BR14] J. Bergner, C. Rezk. *Comparison of models for (∞, n)-categories, II.* arXiv preprint arXiv:1406.4182 (2014).

[Fr72] P. Freyd, *Aspects of topoi.* Bulletin of the Australian Mathematical Society 7.1 (1972): 1-76.

[GJ72] A. Grothendieck, J. L. Verdier. *Théorie des topos (SGA 4, exposés i-vi).* Springer Lecture Notes in Math 269 (1972): 270.

[GK17] D. Gepner, J. Kock. *Univalence in locally cartesian closed ∞-categories*. Forum Mathematicum. Vol. 29. No. 3. 2017.

[Ka15] C. Kapulkin, *Locally cartesian closed quasicategories from type theory.* arXiv preprint arXiv:1507.02648 (2015).

[KL12] C. Kapulkin, P. L. Lumsdaine. *The simplicial model of univalent foundations (after Voevodsky).* arXiv preprint arXiv:1211.2851 (2012).

[La65] W. Lawvere, *An elementary theory of the category of sets*, Proceedings of the National Academy of Science of the U.S.A 52 pp.1506-1511 (1965)

[La70] W. Lawvere. *Quantifiers and Sheaves*. Actes du congres international des mathematiciens, Nice. Vol. 1. 1970.

[Lu09A] J. Lurie, *Higher Topos Theory*, Annals of Mathematics Studies 170, Princeton University Press, Princeton, NJ, 2009, xviii+925 pp. A

[Lu09B] J. Lurie, $(\infty, 2)$-*Categories and the Goodwillie Calculus I.* arXiv preprint arXiv:0905.0462 (2009).

[Lu12] J. Lurie, *Higher algebra*. Preprint, available at www.math.harvard.edu/lurie

[RaA] N. Rasekh, *Yoneda Lemma for Simplicial Spaces*, arXiv preprint arXiv:1711.03160 (2017)

[RaB] N. Rasekh, *Cartesian Fibrations and Representability*, arXiv preprint arXiv:1711.03670 (2017)

[RaC] N. Rasekh, *Complete Segal Objects*, In Preparation

[RaD] N. Rasekh, *A Theory of Elementary Higher Toposes*, In Preparation

[Re01] C. Rezk, *A model for the homotopy theory of homotopy theory*, Trans. Amer. Math.Soc., 353(2001), no. 3, 973-1007.

[Re05] C. Rezk, *Toposes and Homotopy Toposes*, *https://faculty.math.illinois.edu/rezk/homotopy-topos-sketch.pdf*

[Ro12] M. Robalo, *Noncommutative motives I: A universal characterization of the motivic stable homotopy theory of schemes*, arXiv preprint arXiv:1206.3645 (2012).

[Sh17] M. Shulman, *Elementary* $(\infty, 1)$-*Topoi*, Online Blog Post, *https://golem.ph.utexas.edu/category/2017/04/elementary_1t*

[Ti72] M. Tierney *Sheaf Theory and the Continuum Hypothesis*, in F. W. Lawvere (ed.) Toposes, Algebraic Geometry and Logic. Springer Lecture Notes in Mathematics no. 274 Springer Verlag. pp. 13-42.

[Ti73] M. Tierney *Axiomatic Sheaf Theory*, in Proceedings oj CIME Conference on Categories and Commutative Algebm. Varenna, 1971, pp. 249-326. Edizione Cremonese. (Reviewed In Mathematical Reviews 50, 7277.)

[Vo10] V. Voevodsky. *Univalent foundations project.* *http://www.math.ias.edu/vladimir/Site3/Univalent_Foundations_files/univalent_foundations_project.pdf*